ALKENE

81 (c) Unsaturated bond

Dilute Potassium permanganate detects C=C/C≡C.

82 (d) All

Water, HCN, etc., can add to acetylene under suitable conditions.

83. (c) $\underset{CH}{\underset{|||}{CH}} + \underset{AsCl_2}{\underset{|}{Cl}} \xrightarrow{AnhydrousAlCl_3} \underset{CH\ AsCl_2}{\underset{||}{CHCl}}$

Arsenic trichloride → Lewisite (β-Chlorovinyl dichloroarsine)

Lewisite is more poisonous than mustard gas and was used in world war -II.

84. (a) $CH_3-CH=CH_2+Cl_2 \xrightarrow{500^oC} \underset{Cl}{\underset{|}{CH_2}}-CH=CH_2+HCl$

This reaction is called allylic halogenation reaction because halogenation occurs at the allylic position of an alkene

85. (c) $n(CH_2=CH-Cl) \xrightarrow{\text{Polymerization}} (-CH_2-\underset{Cl}{\underset{|}{CH}}-)_n$

(PVC)

86. (c) $CH_2=CH_2+Br_2 \xrightarrow{NaCl} \underset{Br}{\underset{|}{CH_2}}-\underset{Br}{\underset{|}{CH_2}} + \overset{2}{\underset{Cl}{\underset{|}{CH_2}}}-\overset{1}{\underset{Br}{\underset{|}{CH_2}}}$

1,2–dibromo ethane 1–bromo–2–chloro ethane

87. (b) $CH_3-CH=CH_2+HBr \xrightarrow[\text{Markownikoff rule}]{} CH_3-\underset{Br}{\underset{|}{CH}}-CH_3$

2–Bromopropa ne

88. (a) $CH_3-CH=CH_2+HBr \xrightarrow{\text{Peroxide}} CH_3-CH_2-CH_2-Br$

89. (c) $CH_3-\underset{\underset{CH_3}{|}}{C}=CH-CH_3 \xrightarrow[(2)Zn/H_2O]{(1)O_2} \underset{\text{Ketone}}{CH_3COCH_3} + \underset{\text{Aldehyde}}{CHOCH_3}$

90. (c) $R-CH=\underset{\underset{R}{|}}{C}-R \xrightarrow[(2)Zn/H_2O]{(1)O_3} \underset{\text{Aldehyde}}{R-CHO} + \underset{\text{Ketone}}{R_2CO}$

91 (b) C_4H_8

Unsaturated compounds contain C=C or C≡C. C_4H_8 follows general formula of alkenes (C_nH_{2n}), so it is unsaturated.

92. (c) $CH_3-CH_2-OH \xrightarrow[H_2SO_4]{\text{Conc.}} CH_2=CH_2+H_2O$

93. (b) $\underset{\text{Ethyl Chloride}}{CH_3-CH_2-Cl} \xrightarrow{\text{alc. }KOH} \underset{\text{Alkene}}{CH_2=CH_2} + KCl + H_2O$

94. (b) Olefin because double bond is present.

95. (b) In $CH_2 \overset{\pi}{\underset{\sigma}{=}} CH_2$ double bond consist of one σ and one π bond

96. (b) $\underset{\text{Alkylhalide}}{R-CH_2-CH_2-X} \xrightarrow[\text{alc. }KOH]{\text{Elimination}} \underset{\text{Alkene}}{R-CH=CH_2} + \underset{\text{Halogen acid}}{HX}$

97. (a) $CH_3-CH_2-CH_2-CH_3 \xrightarrow[475\ K]{H_2SO_4} \underset{\text{More symmetrical(major product)}}{CH_3-CH=CH-CH_3}$

$CH_3-CH_2-CH_2-CH_3 \xrightarrow[475\ K]{H_2SO_4} \underset{\text{Less symmetrical or unsymmetrical (minor product)}}{CH_2=CH-CH_2-CH_3}$

It is based on Saytzeff's rule. According to this more symmetrical or more alkylated alkene predominates.

98. (b) $CH_3-CH_2-CH_2-OH \xrightarrow{PCl_3} \underset{\text{(a) propylchloride}}{CH_3-CH_2-CH_2-Cl} \xrightarrow{\text{Alc.}KOH} \underset{\text{(b) Propene}}{CH_3-CH=CH_2}$

99. (b) $\underset{n-\text{Propyl bromide}}{CH_3-CH_2-CH_2-Br} + KOH \xrightarrow{C_2H_5OH} \underset{\text{Propene}}{CH_3-CH=CH_2} + KBr + H_2O$

100. (b) $CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_2-Br + KOH \xrightarrow[\text{(alc)}]{} CH_3-\overset{\overset{CH_3}{|}}{C}=CH-CH_3 + KBr + H_2O$

In this reaction 1° carbonium ion is formed which rearranges to form 3° carbonium ion from which base obstruct proton. Hence 2-methyl-2-butene is formed as a main product.

$$\underset{1^o \text{ carbonium less stable}}{CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-\overset{+}{C}H_2} \xrightarrow{\text{Methyl shift}} CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{+}{|}}{C}}-CH_2-CH_3$$

$$\xrightarrow[\text{which is less hydrogenated}]{\text{Elimination of proton from } \beta \text{ carbon}} \underset{\text{2-Methyl-2-Butene}}{CH_3-\overset{\overset{CH_3}{|}}{C}=CH-CH_3}$$

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